

Pearson Edexcel International GCSE

(9-1) ICT: Digital Graphics Masterclass

Graphics software is used to create, edit, and manipulate visual images. These digital graphics are frequently integrated into other ICT products, such as multimedia presentations, desktop published documents, or web pages. Popular application examples include *Adobe Photoshop*, *GIMP* (for bitmap images), *Adobe Illustrator*, *Inkscape*, and *Google Drawings* (for vector graphics).

1. Features of Image Types: Bitmap vs. Vector

All digital graphics fall into one of two primary architectural categories. Understanding their differences is crucial for choosing the correct tool for a given project.

Bitmap Images

Bitmap images (sometimes called raster graphics) are typically used for photographs and scanned documents.

- **Composition:** A bitmap image is made up of a grid of tiny squares called **pixels** (picture elements). Each individual pixel can display only one single color at a time.
- **Resolution:** The quality of a bitmap is determined by its resolution, which refers to the number of pixels packed into a given area. Higher-resolution images contain more pixels, creating a sharper, more detailed picture.
- **Pixelation:** When a bitmap image is enlarged (stretched), the individual pixels become visible as blocky squares, causing the image to lose clarity and look jagged. This effect is known as **pixelation**.

Vector Graphics

Vector graphics are typically used for logos, text typography, icons, and technical illustrations.

- **Composition:** Instead of using pixels, vector graphics are built from mathematical equations that calculate the paths, lines, curves, shapes, and coordinates of objects relative to one another.
- **Scalability:** Because vectors rely on math rather than a fixed grid of squares, they can be scaled or enlarged infinitely without ever losing quality or pixelating. The software simply recalculates the mathematics to redraw the lines perfectly sharp at any size.
- **Realism Limitation:** A major drawback of vector graphics is that they struggle to represent natural objects or complex scenes realistically. They cannot easily capture the subtle color gradients, shadows, and textures found in real-world photographs.

2. Structural Image Composition: Working with Shapes, Text, and Layers

Creating or editing graphics efficiently requires working with structured object assets and organizing them cleanly on your canvas.

Basic Shape Integration

Graphics applications provide vector-based geometric tools to construct compound images by combining primitive assets:

- **Rectangles & Squares:** Form foundational layout containers or solid background borders.
- **Circles & Ovals:** Used for round design elements. Increasing the pixel grid or resolution allows curves to look smooth rather than jagged.
- **Lines, Triangles, and Arrows:** Used extensively to build technical schematics, flowcharts, or user interface components.
- **Text Boxes:** Allow you to insert typography, captions, and labels directly over graphic layouts.

Organizing Layouts with Layers

Advanced graphics software uses a stacked system called **layers** to organize components.

- **Independent Editing:** Each element or asset can be placed on its own transparent layer. This allows you to edit, transform, recolor, or delete one specific part of an image without affecting any of the other surrounding assets.
- **Stacking Order:** Layers can be dragged up or down to change which items appear in front of or behind other elements on the canvas (e.g., bringing a text layer to the absolute front so it sits clearly over a solid background rectangle).

3. Image Editing and Transformation Techniques

Modifying existing graphics involves changing their compositions or erasing flaws using dedicated editing tools.

Aspect Ratio Control

When resizing any image, maintaining its **Aspect Ratio** is vital. The aspect ratio is the proportional relationship between an image's width and its height.

- **Distortion:** If you pull an image's resizing handles from only one side or top edge, you alter its width without scaling its height, causing the image to look stretched, squashed, or **distorted**.
- **Proportional Scaling:** To prevent distortion, always scale images from their corner adjustment handles, or hold down the Shift key while resizing to lock the aspect ratio.

Original [] Distorted (Side Drag) [] Proportional (Corner Drag) []

(Correct)

(Wrong)

(Correct)

Transformation Operations

- **Cropping:** Trimming away unwanted outer edges or background areas of an image to change its canvas boundaries, focus on a specific subject, or alter its layout proportions.
- **Grouping:** Locking multiple individual shapes or layers together so they act as a single, unified object. Grouped items can be moved, resized, or rotated together, preserving their layout spacing.
- **Rotation:** Pivoting an object or an entire canvas around a central axis to change its angle of orientation.
- **Deletion of Unwanted Aspects:** Using clone stamps, erasers, or selection masks to paint over or erase unwanted elements from a photograph (such as removing a background power line or blemish).

4. Legality in Graphics: Understanding Copyright

When sourcing images from the internet to build your digital products, you must respect intellectual property regulations.

- **Copyright Infringement / Breach:** Copying, altering, or publishing an image without obtaining explicit permission or a valid license from the original creator is a breach of copyright law.
- **Safe Sourcing Practices:** To avoid legal breaches, use images that are explicitly marked as *Public Domain*, distributed under *Creative Commons* licensing (which often requires attribution), or sourced from royalty-free stock repositories.

Student Activities

Activity 1: Tracking Pixelation and Resolution

1. Open a bitmap image (such as a .jpeg or .png photograph) inside a web browser or image viewer.
2. Zoom in repeatedly (to 400% or higher) on a sharp boundary line within the photograph.
3. Document your observations: Describe how the line breaks down into small, distinct squares of solid color, and explain how this links to the concept of **pixelation**.

Activity 2: Maintaining Aspect Ratio

1. Insert any circular shape asset into a presentation slide or word processing document.
2. Select the shape, click the handle on the far-right edge, and drag it horizontally outward. Note down how the circle distorts into an oval.
3. Undo your change. Now, click and drag the handle located at the exact bottom-right corner while holding down the **Shift key**.
4. Explain why locking the aspect ratio is critical when scaling image assets in professional presentations and documents.

Chapter Questions

1. Which one of these options must be strictly maintained when resizing a graphic asset to prevent it from becoming visually distorted? A) Z axis B) Aspect ratio C) Rotation D) Grouping

2. Explain why vector graphics do not pixelate or lose quality when they are enlarged to extreme sizes.
3. Describe how layers can be used to construct a complex digital graphic image.
4. State the name of the tiny, single-colored square elements that combine to make up a standard bitmap image.
5. Give one distinct drawback of choosing to use vector graphics instead of bitmap graphics when representing real-world objects.
6. State the term used to describe the action of cutting away the outer edges of an image to remove unwanted background space.
7. Explain what occurs if a digital designer breaches copyright laws when sourcing graphics for a commercial web page.
8. State the feature used to lock multiple separate shapes or text items together so they can be moved or rotated as a single entity.
9. Define the term "resolution" in the context of bitmap photographs.
10. Which type of graphic file format would you choose to preserve a logo's transparent background when exporting it for a website? A) .jpeg B) .bmp C) .png D) .vector

Answer Key & Explanations

1. **Correct Answer: B (Aspect ratio)** *Explanation:* The aspect ratio defines the proportional relationship between width and height. Maintaining it ensures that both axes scale equally, preventing the graphic from stretching or squashing.
2. **Core Explanation:** * Vector graphics do not rely on a fixed grid of colored squares. Instead, they are composed of mathematical equations that calculate paths, lines, and coordinates.
 - When a vector is enlarged, the software recalculates these mathematical formulas instantly, redrawing the lines with perfect sharpness at any size.
3. **Core Layer Description:** * Layers function like a stack of clear, transparent sheets placed on top of one another.
 - Different elements (like backgrounds, shapes, or text labels) are placed on separate layers so they can be edited, moved, or deleted independently without interfering with or altering the rest of the image.
4. **Correct Answer: Pixels (Picture elements)** *Explanation:* Pixels are the smallest individual building blocks of a bitmap graphic, with each pixel holding a single solid color value.
5. **Core Drawback:** * Vector graphics struggle to represent complex photographic realism because their mathematical shapes cannot easily replicate the soft gradients, intricate textures, and realistic shadows found in the natural world.
6. **Correct Answer: Cropping** *Explanation:* Cropping selectively discards outer horizontal or vertical segments of a canvas to refocus the frame on a specific subject area.
7. **Core Legal Outcome:** * Sourcing and publishing copyrighted images without permission constitutes an intellectual property breach, which can lead to legal action, fines, or forcing the organization to take down their website.
8. **Correct Answer: Grouping** *Explanation:* Grouping binds selected components together, allowing transformations to apply uniformly across all items in the group simultaneously.
9. **Core Definition:** * Resolution refers to the density of pixels within a bitmap image. It is often measured in DPI (dots per inch) or total pixel dimensions; a higher resolution means more details and better print quality.
10. **Correct Answer: C (.png)** *Explanation:* The Portable Network Graphics (.png) format

features a dedicated alpha channel that supports background transparency layers, whereas formats like .jpeg force a solid white background box around the graphic.